

Forensics Report Assignment

26 – July - 2024

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1. Executive Summary

1.1 Case & Event Description

Two passengers, Jane Esteban and John Fredricksen, were apprehended by Wellington customs based on intelligence from the Australian government. A search revealed 1 kilogram of methamphetamine hidden in the linings of their bags, leading to their arrest and further investigation. According to intelligence, both had Windows laptops that were taken into custody for further investigations.

During interrogation, Jane disclosed their plan to deliver the bags to Eastbourne Library, with a backup location at 666 Rewera Avenue, Petone, to meet an associate named Steve. John remained silent. Authorities subsequently raided the Petone address, uncovering drugs, firearms, and a desktop computer, though no one was present. The items found implicated Steve as the supplier.

1.2 Brief Methodology

The forensic investigation followed a systematic approach to ensure data integrity. Key steps included securing digital evidence and creating forensic images of suspects' devices to prevent tampering. Industry-standard tools like Autopsy, WinHex, TrueCrypt, and OSForensics were used to analyze emails, documents, media files, communications, and metadata.

Steganography detection and pattern recognition techniques were employed to uncover hidden or deleted files. MD5 hashes were calculated to maintain evidence integrity. Detailed documentation was maintained throughout the process to ensure the evidence's suitability for analysis and potential court proceedings. This approach aimed to establish a clear link between the suspects and their criminal activities.

1.3 Evidence Collection

Digital evidence was meticulously extracted from the suspects' devices using industry-standard forensic tools. The collected data included ISO images and memory dumps from three devices. Customs officers utilized FTK Imager to capture drive and memory images from John Fredricksen's laptop, Jane Esteban's laptop, and a desktop computer at the raided address.

MD5 hashes were calculated to ensure the integrity and authenticity of the images. These preserved images were then provided to the digital forensics team for thorough analysis and investigation of the drug smuggling activities.

1.4 Conclusion

- The digital investigation has provided significant insights into the suspects' involvement in the drug smuggling operation.
- Key connections were identified, linking the suspects, particularly Steve and John, to the illicit activities.
- Critical digital evidence found on the devices corroborated their participation in the drug trafficking scheme.
- Jane was proven to be innocent.

This report outlines the forensic process from start to finish, detailing the methodology and tools used to uncover the illegal activities of John and Steve. It presents a comprehensive analysis of the digital evidence, including metadata and a timeline of events. The findings offer clear evidence of the suspects' roles in the drug smuggling operation and will be crucial for the authorities' case handling.

2. Methodology

2.1 Investigation

The forensic investigation was conducted using a structured and methodical approach to ensure the integrity and reproducibility of the findings. The process involved several key steps:

Securing Digital Evidence: The digital evidence from the suspects' devices was secured to prevent any tampering or loss of data.

Imaging Devices: Forensic images of the devices were created, including ISO images and memory dumps, to allow for a thorough examination without altering the original data.

Forensic Analysis Tools: Various industry-standard forensic tools, such as Autopsy, Winhex, Image Stenography, Truecrypt, Hashcalc, and OSForensics, were utilized to examine the acquired drives and images for suspicious data or files.

Data Examination: The contents of each device were analyzed to identify relevant data, including emails, documents, media files, and communications. Metadata, such as timestamps of creation and file sizes, were examined to gain further insights into the files.

Steganography Detection: Given initial findings that Steve knew how to use steganography, detection techniques were employed to uncover any hidden data within seemingly ordinary files.

Pattern and Connection Analysis: The analysis aimed to uncover patterns, connections, and hidden or deleted files that could reveal the suspects' activities, communications, and plans. Special attention was given to metadata and timestamps to construct an accurate timeline of events and interactions among the suspects.

Integrity Assurance: MD5 hashes of the drive and memory images, as well as all extracted evidence, were calculated before and after the investigation to ensure the data's integrity. This step confirmed that no modifications occurred during the process.

Documentation and Reporting: All findings were meticulously documented to maintain the integrity of the evidence, ensuring it was suitable for analysis and potential court proceedings.

2.2 Use of Forensics Software:

The forensic investigation utilized various tools and additional necessary software to extract valuable evidence, ensuring thorough and accurate analysis. The tools used included:

Autopsy v-4.21.0

Purpose: Conduct a forensic investigation of the suspects' virtual images.

Functionality: Autopsy was used to add the image as a data source for a new case. Its powerful variety of tools and ingest modules allowed for a thorough exploration of file systems, extraction and analysis of metadata, recovery of deleted files, and detection of user behavior. Keywords were searched to obtain relevant data.

OSForensics v-11.0

Purpose: Analyze Steve's and other suspects virtual image comprehensively.

Functionality: OSForensics' advanced features were used to scan and index the file system, search for specific file types, examine metadata, and recover deleted or hidden files. Its powerful search capabilities identified and extracted critical evidence by methodically analyzing the data and correlating it with the investigation objectives.

WinHex v-21.2

Purpose: Thorough examination of memory dumps from the suspects' virtual images.

Functionality: Loaded memory dump files and used WinHex's powerful hex editor to navigate through raw data. Analyzed memory contents, active processes, loaded modules, and network connections at the time of memory capture. Searched for specific keywords, file signatures, and other indicators of suspicious activity.

Image Steganography v-1.5.2.0

Purpose: Detect and decode hidden data within images.

Functionality: It was used in this investigation because John and Steve were found to have hidden secret information in images. This tool was specifically used to conceal a hidden image containing crucial details about their operation.

TrueCrypt v-7.2

Purpose: Decrypt encrypted files and drives.

Functionality: A very famous encryption software for volumes and files, it was used by the two suspects, John and Steve, to encrypt an important document regarding their drug operation.

MD5 Calculator

Purpose: Ensure the integrity of files and evidence.

Functionality: Calculated MD5 hashes of files, ISO images, and memory dumps to check for any modifications during the investigation.

By utilizing these tools and techniques, the forensic team ensured a thorough and accurate analysis, maintaining the integrity of the evidence for potential court proceedings.

The obtained drive images of the three suspects, Jane Esteban, John Fredricksen, and Steve Kowhai, were examined in depth using Autopsy and OSForensics. This analysis aimed to extract relevant data artifacts such as web history, file metadata, images, logs, chats, and emails to understand the activities carried out on their devices.

2.2.1 Proton mail Analysis

The email accounts of the suspects, primarily using ProtonMail, were examined using credentials acquired from their drive images. The emails exchanged between the three suspects were scrutinized to identify any suspicious documents and understand their network of communication.

2.2.2 Discord Chat Log Analysis

Discord chat logs were extracted and analyzed using Autopsy. The focus was on conversations between Jane Esteban, John Fredricksen, and Steve Kowhai, which revealed details of the smuggling operation, including their plans and methods.

2.2.3 Keyword Analysis

Autopsy, along with Magnet Axiom, was employed for keyword searches to quickly identify documents, emails, and other files containing relevant information about the suspects' activities. Keywords such as "methamphetamine," "meth," "drugs," "password," "routes," "New Zealand," and more were used to uncover evidence of drug-related activities. Additionally, terms like "Protonmail," "encrypted," and "steganography" were used to find communication methods and relevant conversations about their plans and methods.

2.2.4 Image Stenography Analysis

Images on the devices were analyzed for hidden data using the Image steganography tool. This tool was crucial for uncovering concealed documents within image files, as Steve Kowhai used it to hide important information. Steve sent vital instructions to John hidden within seemingly ordinary images.

2.2.5 Data Artifacts

Data artifacts such as web browser history, cache, and download records were extracted through Autopsy to gather evidence on the suspects' online activities and interactions. This provided insights into the research and preparation done for the smuggling operation, revealing web searches, browsed websites, and files downloaded by the suspects.

2.3 Hash Table Calculation

File	Hash at beginning of investigation	Hash at end of investigation
Narcos -1	996182c381ec9e7025f40519107615e4	996182c381ec9e7025f40519107615e4
Narcos -2	56823dee9b24a40407bec184f80261c2	56823dee9b24a40407bec184f80261c2
Narcos -3	ce707bf783dde13ed42196cd6e47308 3	ce707bf783dde13ed42196cd6e47308 3
Narcos -Mem- 1	3469089f9a26b0b51a4ee985cd1c3008	3469089f9a26b0b51a4ee985cd1c3008
Narcos -Mem- 2	fd7027c7bffedd653226355467c2b1ef	fd7027c7bffedd653226355467c2b1ef

Narcos -Mem- 3	7c155dae658fb059586b3ab5144e21d 2	7c155dae658fb059586b3ab5144e21d 2
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3. Findings

The evidence findings from the three suspects are detailed below, beginning with inculpatory evidence followed by exculpatory evidence.

Suspect	Username	Laptop Name	Operating system	OS Build number
Steve Kowhai	Steve	SK-DESKTOP	Windows 10 pro	16299
John Fredrickson	JohnF	JOHNFLAPTOP1	Windows 10 pro	16299
Jane Estaban	JaneE	JELAPTOP	Windows 10 pro	17763

3.1 Finding 1 – John’s online activities and interactions

Path:Users/JohnF/AppData/Roaming/Mozilla/Firefox/Profiles/fj248tbu.default/places.sqlite
Calculated Hash: 50a698fff857f00065e7b6b0ceee2188

3.1.1 Discussion

John's web activity reveals a deliberate and methodical approach to drug smuggling. His searches included terms such as "encryption methods YouTube," "detector dogs," "body packing," and "concealment." This indicates a concerted effort to evade authorities and ensure the smooth operation of his illegal activities.

His extensive research into hiding drugs and circumventing detection, including methods to bypass New Zealand's drug detection dogs, highlights his familiarity with drug trafficking and proactive steps to avoid capture. Additionally, his interest in encryption techniques and anti-virus software suggests a desire for secure information exchange.

The following evidence will show that these activities are inculpatory, demonstrating John's intent and calculated efforts to engage in and conceal his drug smuggling operations.

3.1.2 Supporting Evidence

Web Search	
Term:	encryption methods youtube
Time:	2019-01-30 05:49:46 GST
Domain:	google.com
Program Name:	FireFox Analyzer
Source	
Host:	Narcos-2_1 Host
Data Source:	Narcos-2
File:	/img_Narcos-2/vol_vol7/Users/JohnF/AppData/Roaming/Mozilla/Firefox/Profiles/fj248tbu.default/places.sqlite

Fig. 1 Web Search – “Encryption Methods Youtube”

Web Search	
Term:	drug detector dogs
Time:	2019-01-31 05:59:46 GST
Domain:	google.com
Program Name:	FireFox Analyzer
Source	
Host:	Narcos-2_1 Host
Data Source:	Narcos-2
File:	/img_Narcos-2/vol_vol7/Users/JohnF/AppData/Roaming/Mozilla/Firefox/Profiles/fj248tbu.default/places.sqlite

Fig. 2 Web Search – “Drug Detector Dogs”

Web Search	
Term:	suitcase concealments for drugs
Time:	2019-01-31 06:12:25 GST
Domain:	google.com
Program Name:	FireFox Analyzer
Source	
Host:	Narcos-2_1 Host
Data Source:	Narcos-2
File:	/img_Narcos-2/vol_vol7/Users/JohnF/AppData/Roaming/Mozilla/Firefox/Profiles/fj248tbu.default/places.sqlite

Fig. 3 Web Search – “Suitcase concealments for drugs”

Web Search	
Term:	body packing
Time:	2019-01-31 06:01:30 GST
Domain:	google.com
Program Name:	FireFox Analyzer
Source	
Host:	Narcos-2_1 Host
Data Source:	Narcos-2
File:	/img_Narcos-2/vol_vol7/Users/JohnF/AppData/Roaming/Mozilla/Firefox/Profiles/fj248tbu.default/places.sqlite

Fig. 4 Web Search – “Body Packing”

Visit Details	
Title:	Cryptographic Algorithms - YouTube
Date Accessed:	2019-01-30 05:54:50 GST
Domain:	youtube.com
URL:	https://www.youtube.com/watch?v=VUaxnqALUI4
Program Name:	FireFox Analyzer
Source	
Host:	Narcos-2_1 Host
Data Source:	Narcos-2
File:	/img_Narcos-2/vol_vol7/Users/JohnF/AppData/Roaming/Mozilla/Firefox/Profiles/fj248tbu.default/places.sqlite

Fig. 5 Web Search – “Cryptographic Algorithms Youtube”

Visit Details	
Title:	Internal concealment of drugs of abuse (body packing) - UpToDate
Date Accessed:	2019-01-31 06:02:21 GST
Domain:	uptodate.com
URL:	https://www.uptodate.com/contents/internal-concealment-of-drugs-of-abuse-body-packing
Program Name:	FireFox Analyzer
Source	
Host:	Narcos-2_1 Host
Data Source:	Narcos-2
File:	/img_Narcos-2/vol_vol7/Users/JohnF/AppData/Roaming/Mozilla/Firefox/Profiles/fj248tbu.default/places.sqlite

Fig. 6 Web Search – “internal Concealment of drugs of abuse (body packing) - UpToDate”

Web Search	
Term:	badiu anti-virus
Time:	2019-01-29 00:15:55 GST
Domain:	google.com
Program Name:	FireFox Analyzer
Source	
Host:	Narcos-2_1 Host
Data Source:	Narcos-2
File:	/img_Narcos-2/vol_vol7/Users/JohnF/AppData/Roaming/Mozilla/Firefox/Profiles/fj248tbu.default/places.sqlite

Fig. 7 Web Search – “badiu anti-virus”

Visit Details	
Title:	NZ's Leading Drug & Explosives Detector Dog Services New Zealand Detector Dog Services Ltd
Date Accessed:	2019-01-31 05:59:50 GST
Domain:	nz-detector-dogs.co.nz
URL:	https://nz-detector-dogs.co.nz/?gclid=EAlalQobChMI-oPvjPaW4AlVhxmPCh3Bmg5wEAAYASAAEgLVz_D_BwE
Referrer URL:	http://nz-detector-dogs.co.nz/?gclid=EAlalQobChMI-oPvjPaW4AlVhxmPCh3Bmg5wEAAYASAAEgLVz_D_BwE
Program Name:	FireFox Analyzer
Source	
Host:	Narcos-2_1 Host
Data Source:	Narcos-2
File:	/img_Narcos-2/vol_vol7/Users/JohnF/AppData/Roaming/Mozilla/Firefox/Profiles/fj248tbu.default/places.sqlite

Fig. 8 Web Search – “New Zealand Detector Dog Services Ltd”

3.2 Finding 2 – Steve’s online activities and interactions

Path: /Users/Steve/AppData/Local/Microsoft/Windows/WebCache/WebCacheV01.dat

Calculated Hash: d69b6ea8e6a1c05c57c4b0523bb37ac0

3.2.1 Discussion

The examination of Steve’s files revealed numerous web searches related to drug dealing and smuggling activities. Notable searches included “international drug routes,” "how to launder money," "drug routes in Wellington," and "what is methamphetamine," indicating significant planning and preparation for his operations.

These findings are crucial to the case, as they connect Steve’s online activities to his smuggling operations. The searches and downloads, including a steganography tool and encryption software, show Steve's intent and efforts to facilitate and conceal his illegal activities, both before and after his interactions with John.

The following evidence will further substantiate these activities as inculpatory, highlighting Steve's involvement in criminal operations.

3.2.2 Supporting Evidence

Web Search	
Term:	how to launder money
Time:	2019-01-28 23:03:51 GST
Domain:	google.com
Program Name:	Microsoft Edge Analyzer
Source	
Host:	Narcos-1_1 Host
Data Source:	Narcos-1
File:	/img_Narcos-1/vol_vol7/Users/Steve/AppData/Local/Microsoft/Windows/WebCache/WebCacheV01.dat

Fig. 9 Web Search – “How to launder money”

Web Search	
Term:	drug routes in wellington
Time:	2019-01-29 01:01:58 GST
Domain:	bing.com
Program Name:	Microsoft Edge Analyzer
Source	
Host:	Narcos-1_1 Host
Data Source:	Narcos-1
File:	/img_Narcos-1/vol_vol7/Users/Steve/AppData/Local/Microsoft/Windows/WebCache/WebCacheV01.dat

Fig. 10 Web Search – “Drug routes in Wellington”

Web Search	
Term:	international drug routes
Time:	2019-01-29 01:04:16 GST
Domain:	google.co.nz
Program Name:	Microsoft Edge Analyzer
Source	
Host:	Narcos-1_1 Host
Data Source:	Narcos-1
File:	/img_Narcos-1/vol_vol7/Users/Steve/AppData/Local/Microsoft/Windows/WebCache/WebCacheV01.dat

Fig. 11 Web Search – “International drug routes”

Visit Details	
Title:	TrueCrypt download starting...
Username:	Default
Date Accessed:	2019-01-30 03:27:28 GST
Domain:	softpedia.com
URL:	https://www.softpedia.com/dyn-postdownload.php/9196da93f480ba08f95035b293f60971/5c50ef6a/44cf/4/2
Referrer URL:	https://www.softpedia.com/dyn-postdownload.php/9196da93f480ba08f95035b293f60971/5c50ef6a/44cf/4/2
Program Name:	Google Chrome
Source	
Host:	Narcos-1_1 Host
Data Source:	Narcos-1
File:	/img_Narcos-1/vol_vol7/Users/Steve/AppData/Local/Google/Chrome/User Data/Default/History

Fig. 12 Web Download – “TrueCrypt”

3.3 Finding 3- Evidence of Established Drug Distribution Network

Path:/Users/JohnF/Documents/Business/clients.ods

Calculated Hash: db804b47095901e20d6666b1e9a56003

Modified: 2019-01-29 08:02:47 GST

Accessed: 2019-01-30 04:04:14 GST

Created: 2019-01-29 00:26:09 GST

Changed: 2019-01-29 08:02:47 GST

3.3.1 Discussion

A spreadsheet found in John's files lists various clients, including their names, locations, drug products, quantities, and delivery frequencies. This document reveals that John operated an extensive and well-organized drug distribution network with numerous regular clients even before his collaboration with Steve Kowhai.

The inclusion of Steve as a new client indicates John's expanding reach and active recruitment efforts. This evidence is crucial as it demonstrates John's significant involvement in a sophisticated international drug operation.

The following evidence will further support this as inculpatory, reinforcing John's key role in the illicit drug trade.

3.3.2 Supporting Evidence

1	Name	Location	Product	Amount	Delivery
2	Ricky Ross	Los Angeles	Mama Coca	20kg	Monthly
3	Frank Lucas	New York, USA	Ferry Dust	15kg	Quarterly
4	Chris Coke	Kingston Jamaica	Coke	20kg	Monthly
5	Steve Kowhai	Wellington, New Zealand	Crank	15kg	Monthly
6	Don Cholito	Puerto Rico	Snow	25kg	Quarterly
7	Manuel Noriega	Panama	Smack	15kg	Monthly
8	Joaquin Guzman	Guadalajara, Mexico	China White	15kg	Monthly
9	Leroy Barnes	New York, USA	Load pack	15kg	Quarterly
10	AL Capone	Siciliy, Italy	Silly putty	25kg	Monthly
11	Jane Esteban	Brisbane, Austraila	Uppers	1 gram	On demand
12	Pablo Esobar	Colombia	White horse	15kg	Quarterly
13	Franz Sanchez	Isthmus City	Mary Jane	20kg	Quarterly
14	Jake Heke	Auckland	Tweak	10kg	Monthly

Fig. 13 clients.ods – “John’s Client List”

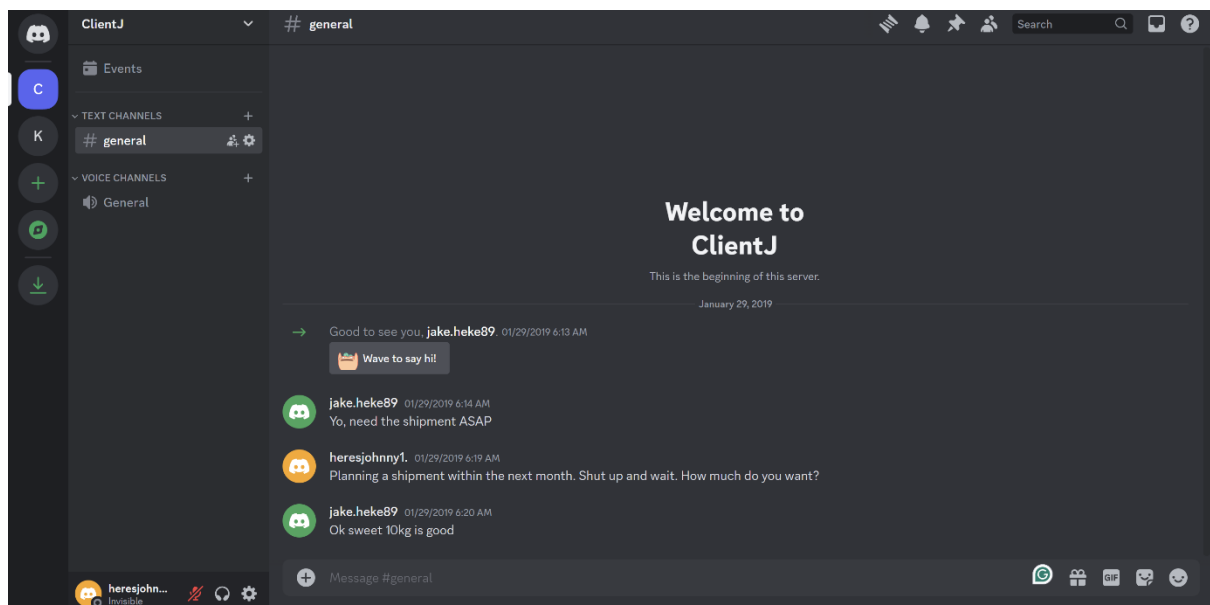


Fig. 14 Discord Logs – “ClientJ”

Track this shipment via the DHL Web Site : <http://www.dhl.com>

Shipment Air Waybill
(Non negotiable)

ORIGIN: B N E
DESTINATION CODE: A K L

1 Payer account number and insurance details
Charge to ☒ Shipper ☐ Receiver ☐ 3rd party ☒ Cash ☒ Cheque ☒ Credit Card
Payer Account No. 001-158545-85
Shipment Insurance see reverse
☒ Yes Insured value (in local currency) 0 Not all payment options are available in all countries.

2 From (Shipper)
Shipper's account number 258-85695 Contact name Johnny Fredrick
Shipper's reference (up to 32 characters but only first 12 will be shown on invoice) AB-20071223-589X
Company name High As a Kite LLC
Address 8515 Haven Wood Trail
Inala, Brisbane
QLD 4077
Australia
Postcode/Zip Code (required) QLD 4077 Phone, Fax or E-mail (required) +1 258 585 965

3 To (Receiver)
Company name
Delivery address DHL cannot deliver to a PO Box
5/34 Hapua Street
Remuera
Auckland 1050
New Zealand
Postcode/Zip Code (required) 1050 Country New Zealand
Contact person Jake Heke Phone, Fax or E-mail (required) +6402145365477

4 Shipment details
Total number of packages 1 Total Weight 20kg
Dimensions in cm: Length 500mm, Width 500mm, Height 600mm
Pieces 575 @ 500mm x 500mm x 600mm

5 Full description of contents
Give content and quantity
1x Pressure cooker
3x Pots
1x Bread Maker

6 Non-Document Shipments Only (Customs Requirement)
Attach the original and four copies of a Proforma or Commercial invoice
Shipper's VAT/GST number Receiver's VAT/GST or Shipper's EIN/SSN
Declared Value for Customs (as on commercial/proforma invoice) Harmonised Commodity Code if applicable
TYPE OF EXPORT ☒ Permanent ☐ Repair / Return ☐ Temporary
Destination duties/taxes If left blank receiver pays duties/taxes
☒ Receiver ☐ Shipper ☐ Other specify approved account number

7 Shipper's agreement (Signature required)
Unless otherwise agreed in writing, I/we agree that DHL Terms and Conditions of Carriage are all the terms of the contract between me/us and DHL and (1) such Terms and Conditions and, where applicable, the Warsaw Convention limits and/or excludes DHL's liability for loss, damage or delay and (2) this shipment does not contain cash or dangerous goods (see reverse).
Signature Johnny Fredrick Date 29 / 01 / 2019

8 Services
Domestic ☐ International ☐ International Document ☐ Express Mail ☐
Not all services are available to and from all locations
☐ Express 9 (to 36 to the USA)
☐ Express 12
☐ Express / Worldwide
☐ Express Envelope
☐ Other
Optional Services (extra charges may apply)
☐ Saturday Delivery ☐ Special Pick-Up
☐ Delivery Notification
☐ Other
DHL Critical Mail ☐ GMS Priority ☐ GMS Standard ☐ Other
DIMENSIONAL/CHARGEABLE WEIGHT
kg * gr
CHARGES
Services
Other
Insurance
VAT
CURRENCY TOTAL
TRANSPORT COLLECT STICKER No.
PAYMENT DETAILS (Cheque, Card No.)
No. :
Type Expires
Picked up by
Route No.
Time Date

For more information see DHL Web Site or call your local DHL office

PARAGOV - EULEXEXPORTOR - 12/19750 VERSION 1 3rd proof

Origin copy

Fig. 15 DHL Shipment – “Shipment to ClientJ”

As shown in the Clients List (Fig. 13), John was contacted by Jake Heke (Client no. 14 (Fig. 13)). The chats are shown in Discord Logs (Fig. 14) regarding Jake requesting the shipment of 10KG Tweak. The DHL Shipment (Fig. 15) shows the record of the shipment and the address of Jake providing evidence of John being an established drug dealer.

3.4 Finding 4 – Image Stenography Communication (Steve & John)

FILENAME: BNE.PNG

Path: /Users/Steve/Downloads/Misc/BNE.png

Calculated Hash: dffa98bdd7371d1806adb09db27ba283

Created: 2019-02-01 04:13:19 GST

FILENAME: PACKAGE.JPG

Path:/Users/JohnF/AppData/Local/Temp/vmware-JohnF/VMwareDnD/770cd05f/package.jpg

Calculated Hash:

df319b6bc0108d9b1e781895a907eeb621a87fff24f83138de6c67d4f646ae5b

Created: 2019-02-01 03:02:24 GST

3.4.1 Discussion

Steganography was detected in an image sent by John to Steve. The image, which appeared to be a city landscape, was decoded using the "Image Steganography" tool with the password "Elchapo2" provided in an email. The decoded image revealed a photograph of a package that John intended to smuggle into New Zealand for delivery to Steve. This finding underscores their use of steganography to conceal their drug smuggling plans and activities.

Further analysis revealed that John and Steve used ProtonMail, an encrypted email service, to exchange this image, named BNE.png. The hidden content within the image detailed information about their operation and potential future deals. Accessing their ProtonMail accounts confirmed the use of secure communication methods to share these clandestine details.

The following evidence will further substantiate these findings as inculpatory, illustrating their methods of evasion and their active involvement in the drug smuggling operation.

3.4.2 Supporting Evidence

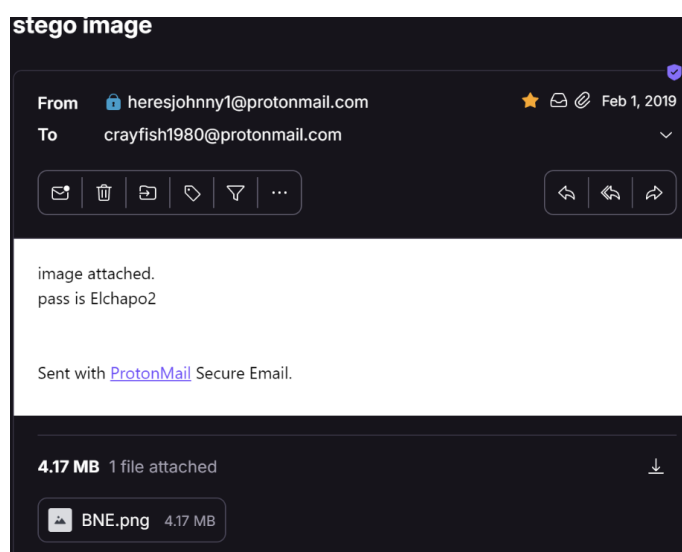


Fig. 16 ProtonMail – “BNE.png Password”



Fig. 16 BNE.png – “Encrypted image”



Fig. 17 Package.jpg – “Actual image”

3.5 Finding 5 – Document Stenography Communication (Steve & John)

Path: /Users/JohnF/Downloads/Attachments-Important, crucial to our method/secret

Calculated Hash: 6c1b3daeda50ba945a76e1bf95ae9012

3.5.1 Discussion

An encrypted document titled “secret.txt” was found in an email sent by Steve to John. The file, encrypted using TrueCrypt, was decrypted after obtaining the password and using the TrueCrypt software. The decrypted document, named “Memo Things,” provided detailed information about John and Steve’s drug smuggling plans.

This document is significant as it reveals the strategic planning and operational details of their drug trafficking business. It provides concrete evidence of their coordination and intentions to expand their illicit activities. The following evidence will further support these findings as inculpatory.

3.5.2 Supporting Evidence

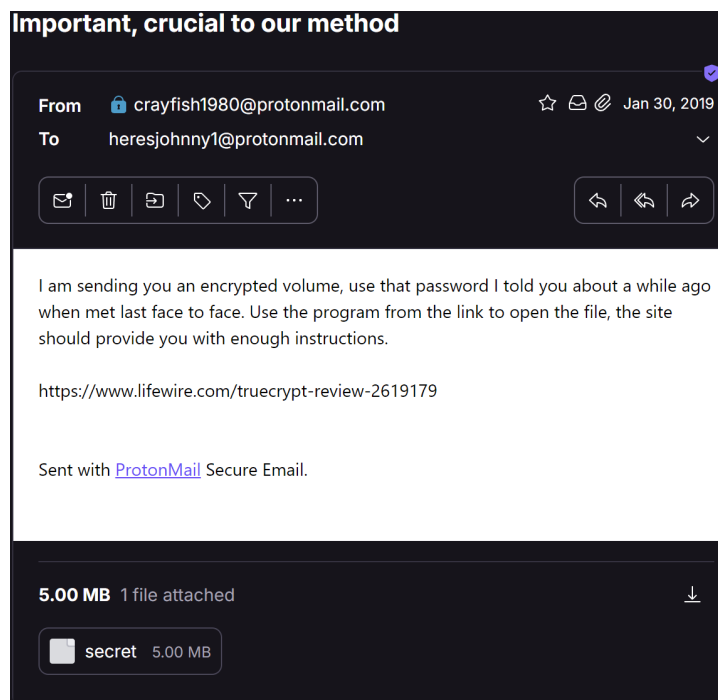


Fig. 18 secret.txt – “Encrypted Document”

Crystal Method

Destination – Wellington

Product – 1 kg crystal

Source – Brisbane (JF)

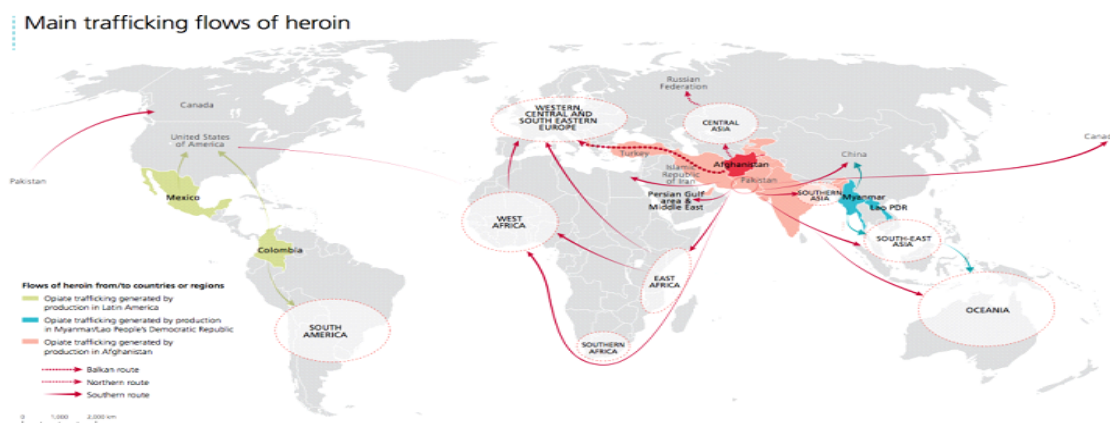
Wellington is part of the lower north island of New Zealand. It is accessible by road and sea. As tempting as the sea is, be aware that the coast of NZ may seem ideal, but everyone knows it's vulnerable as there have been several news reports about it.

The product is coming from Brisbane, Australia and as a result, will need to be delivered by sea or plane. Shipping the product would be ideal, the risk of detection is lower but would result in higher costs for a small amount of product. The product being delivered is only a test to check it's quality and will be the deciding factor in any decision to continue doing business with you. If the product is up to scratch, I will buy a larger supply next time. If we continue to do business it might be more practical to use shipping for future transactions.

This consignment will be delivered by plane, which will decrease travel time and lower the cost for all parties involved. Delivery by plane will bring along with it its own complications such as concealment, security screening/checks and drug dogs. I trust that you are aware of the risks and will take all necessary precautions.

Below is a map of drug flows from countries, could give us some info for later trades and where we can get more product from in the future.

Fig. 19 Memo Things – “Drug Planning”



MAP 2 : Interregional trafficking flows of methamphetamine, 2011-2014

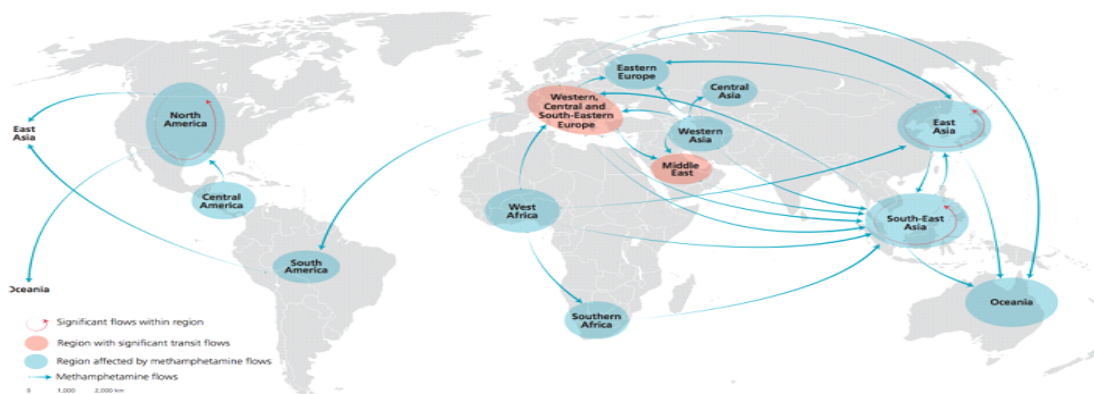


Fig. 20 Memo Things – “Secret Document Drug Routes”

3.6 Finding 6 – Discord Chat Communications (Steve & John)

LOG NAME: 000006.log

Path: /Users/Steve/AppData/Roaming/Discord/Local Storage/leveldb/000006.log

Calculated Hash: 19da7fc1b8da010f787b4671aad2570f

Created: 2019-01-29 07:43:06 GSTs

IMAGE NAME: FLIGHTBOOKINGS.PNG

Path: /Users/Steve/Documents/Misc/flightbookings.PNG

Calculated Hash: d1b21a1cddcb3494d637f2424cc5f0f1

Created: 2019-02-02 06:28:44 GST

3.6.1 Discussion

The analysis of Discord chat logs from Steve and John's accounts provides critical evidence of their involvement in the drug trafficking operation. The chat logs detail plans for John to deliver a shipment of drugs to Steve in New Zealand and discussions about employing steganography to conceal a document within a seemingly ordinary image.

The communications reveal their planning, including the use of steganography to hide sensitive information and their coordination regarding delivery quantities and new suppliers. The final messages arrange a meeting in Wellington for the drug transaction, underscoring their deliberate efforts to conceal their illegal activities along with an image of John's flight ticket from Brisbane to Wellington.

This evidence is crucial as it confirms their conspiracy and methodical approach to drug smuggling. The following evidence will further substantiate these findings as inculpatory.

3.6.2 Supporting Evidence

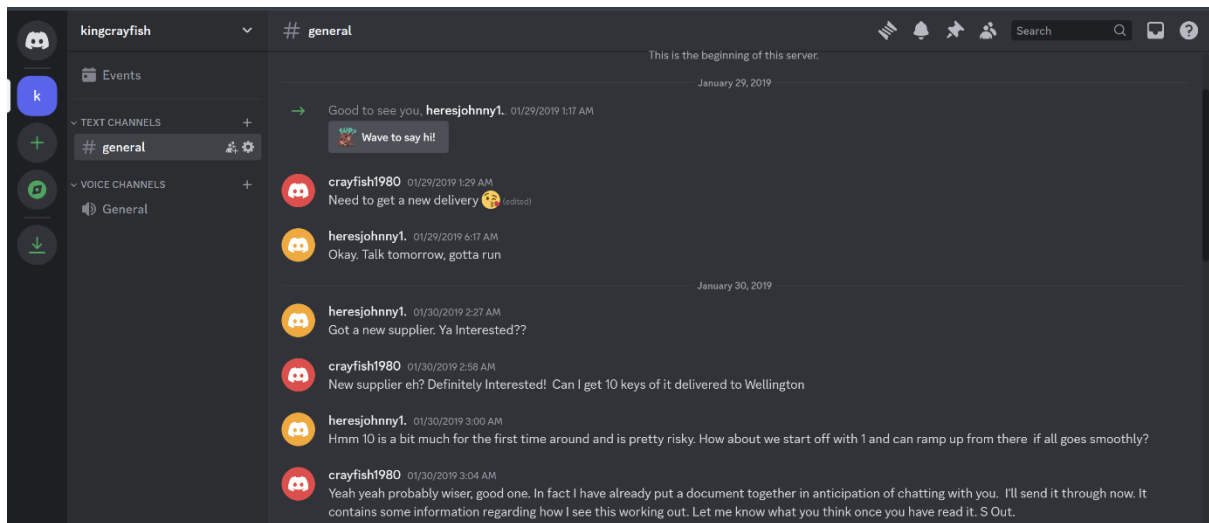


Fig. 21 Discord Chats (1/3) – “Steve & John Discord Communication”

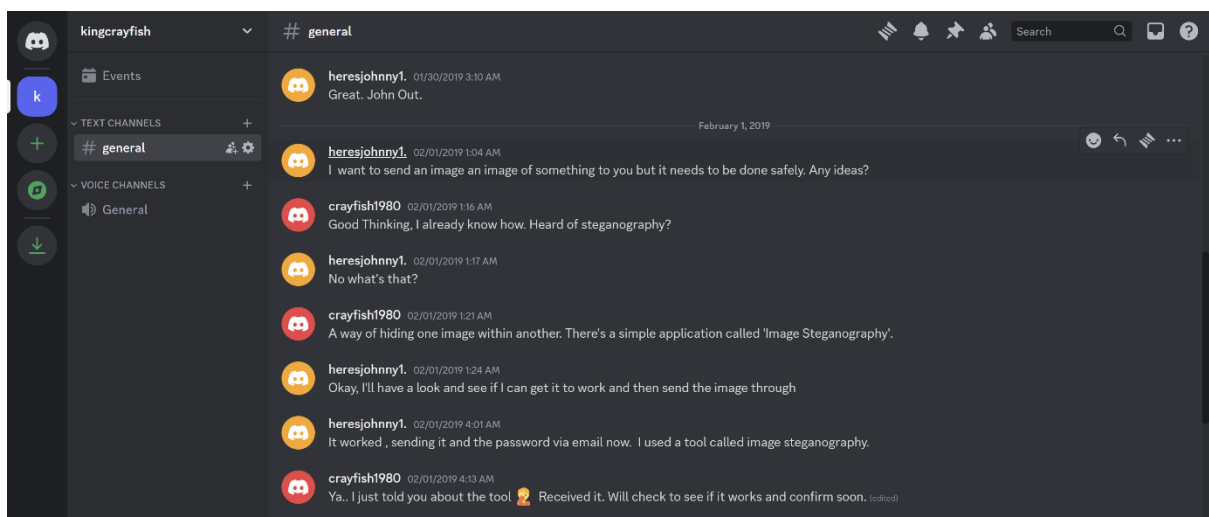


Fig. 22 Discord Chats (2/3) – “Steve & John Discord Communication”

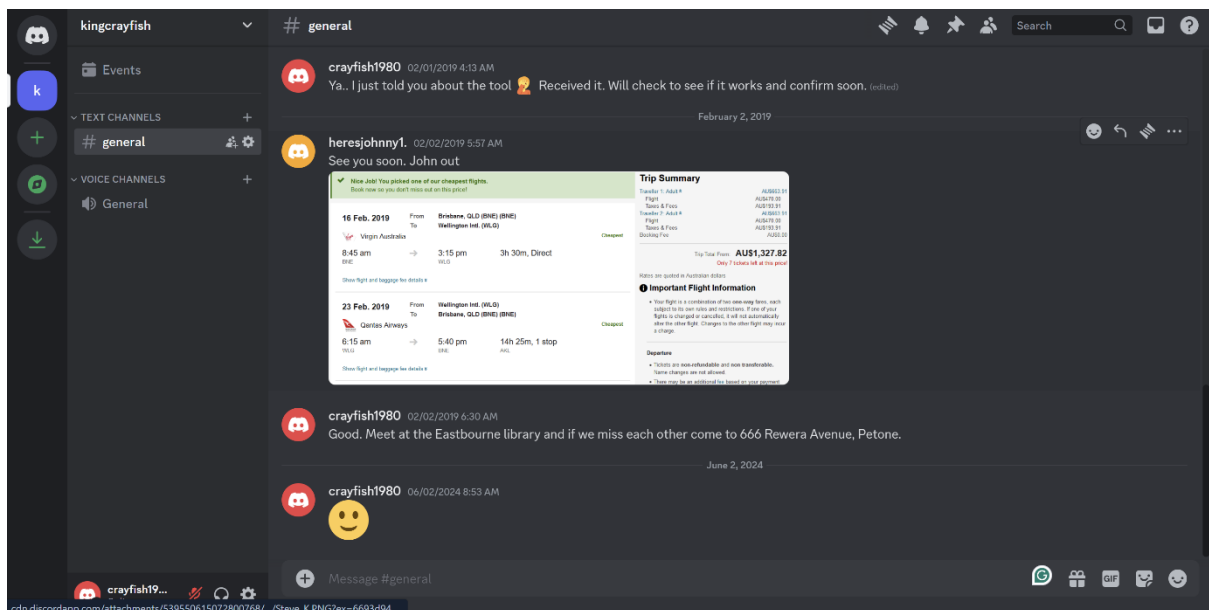


Fig. 23 Discord Chats (3/3) – “Steve & John Discord Communication”

Nice Job! You picked one of our cheapest flights.
Book now so you don't miss out on this price!

Date	From	To	Flight	Time	Duration	Class
16 Feb. 2019	Brisbane, QLD (BNE) (BNE)	Wellington Intl. (WLG)	Virgin Australia	8:45 am	3:15 pm	Cheapest
					3h 30m, Direct	
23 Feb. 2019	Wellington Intl. (WLG)	Brisbane, QLD (BNE) (BNE)	Qantas Airways	6:15 am	5:40 pm	Cheapest
					14h 25m, 1 stop	

Trip Summary

Traveller	Flight	Taxes & Fees	Booking Fee
Traveller 1: Adult	AUS\$663.91	AUS\$470.00	AUS\$193.91
Traveller 2: Adult	AUS\$663.91	AUS\$470.00	AUS\$193.91
Booking Fee			AUS\$0.00

Trip Total From: AU\$1,327.82
Only 7 tickets left at this price!

Important Flight Information

- Your flight is a combination of two one-way fares, each subject to its own rules and restrictions. If one of your flights is changed or cancelled, it will not automatically alter the other flight. Changes to the other flight may incur a charge.
- Tickets are non-refundable and non-transferable. Name changes are not allowed.
- There may be an additional fee based on your payment.

Fig. 24 John’s Airline Ticket – “Brisbane to Wellington”

3.7 Finding 7 – Drug Samples

Path: /\$Recycle.Bin/S-1-5-21-1474204758-2504895174-1356074821-1001/\$RA3IE5E.jpg

Calculated Hash: 480da4ff5734c6d20787af08fc9da46e

Modified: 2019-01-31 06:58:22 GST

Accessed: 2019-01-31 07:04:13 GST

Created: 2019-01-31 06:58:22 GST

Changed: 2019-02-01 06:48:41 GST

Path: /\$Recycle.Bin/S-1-5-21-1474204758-2504895174-1356074821-1001/\$RIIK1AS.jpg

Calculated Hash: 60398a8ec59753ce5aadb99a566844d7

Created: 2019-01-31 06:57:04 GST

3.7.1 Discussion

In Steve's files, images were discovered depicting various drugs associated with his smuggling operations. These images included both general drug types used in his trafficking activities and specific samples of the substance intended for smuggling into New Zealand through John.

This evidence is crucial in establishing Steve's active participation in drug trafficking and his strategic planning for smuggling operations. The following evidence will further reinforce these findings as inculpatory.

3.7.2 Supporting Evidence



Fig. 25 Drug Samples - "Drug Image (1/2)"



Fig. 26 Drug Samples – “Drug Image (2/2)”

3.8 Finding 8 – Drug meetup location (Steve & John)

Path: /Users/Steve/Documents/Misc/airport crystals.jpg

Calculated Hash: f938f92d9dbaeda2e92246deeebe830a

Created: 2019-01-31 01:25:18 GST

Path: /Users/Steve/Documents/Misc/dropoff.jpg

Calculated Hash: 7feb763bebe42c3a3464c9ee3bb1d3e8

Created: 2019-02-02 05:06:05 GST

3.8.1 Discussion

Google Maps images were recovered from the files of Steve and John, detailing the route and providing directions to their agreed meeting location, identified as the “Eastbourne Library.” These images corroborate the logistics discussed in their Discord communications regarding their planned rendezvous.

The maps not only confirm the specific meeting point but also demonstrate the careful planning involved in their drug trafficking operations.

This finding is pivotal in linking the digital correspondence with physical locations, further solidifying the inculpatory nature of their activities.

3.8.2 Supporting Evidence

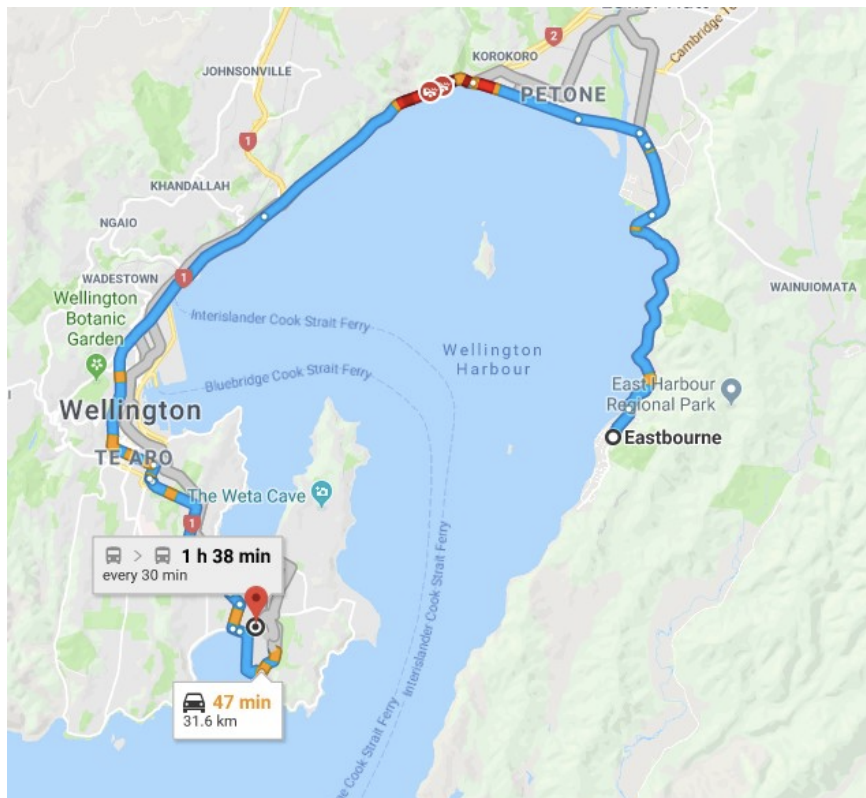


Fig. 27 Google Maps – “Meetup Location”

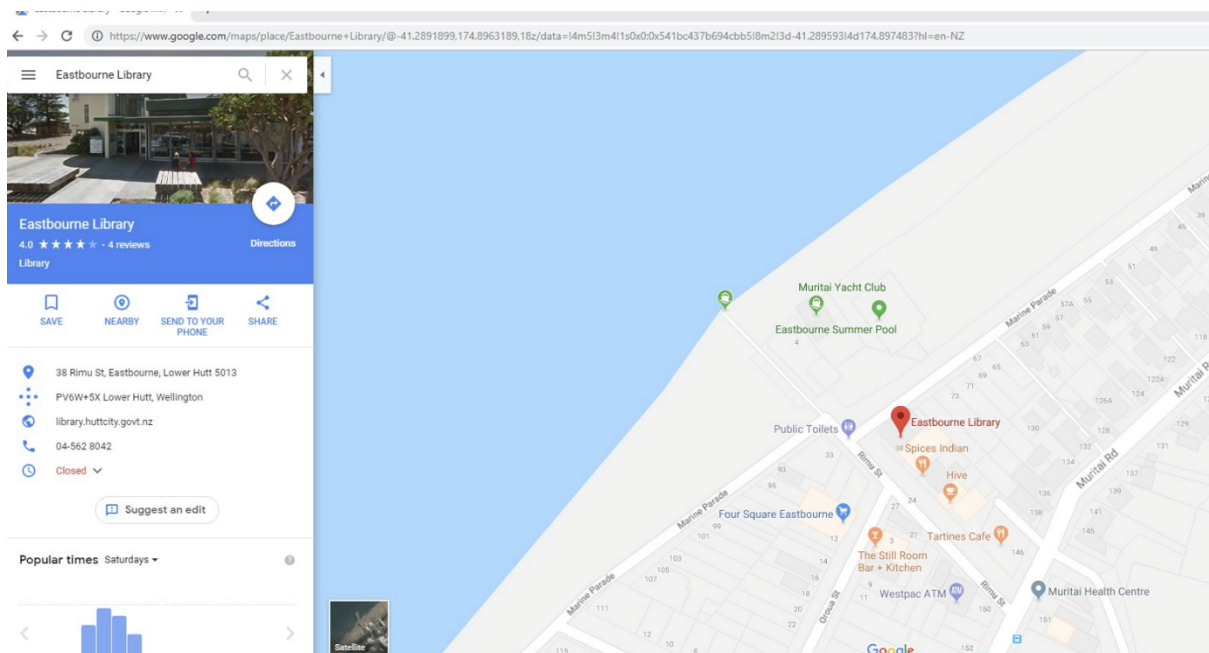


Fig. 28 Google Maps – “Eastbourne Library”

3.9 Finding 9 – Jane’s online activities and interactions

Path: /Users/JaneE/AppData/Local/Microsoft/Windows/WebCache/WebCacheV01.dat

CALCULATED HASH: 1911bbd11dc5c49efd4d1ad37263fab3

3.9.1 Discussion

A detailed analysis of Jane's online activities affirms her allegiance to the ABI and provides exculpatory evidence supporting her innocence. Jane's web searches included terms such as “undercover cop survival,” “how to act like a meth addict,” “hidden small cameras,” “quasar rat,” “voice recorders,” and “Remote Access Trojan.”

These searches indicate that Jane was actively developing her undercover persona to convincingly portray a drug addict. Her research on methamphetamine and related survival tactics, along with the use of spying devices, was aimed at gaining John's trust to gather critical information about his and Steve's activities.

The evidence demonstrates that Jane's actions were in alignment with her duties as an undercover ABI agent, confirming her commitment to infiltrating John's drug operations and collecting intelligence for law enforcement. This evidence is crucial in proving her innocence.

3.9.2 Supporting Evidence

Web Search	
Term:	how to act like a meth addict
Time:	2019-01-29 00:08:21 GST
Domain:	bing.com
Program Name:	Microsoft Edge Analyzer
Source	
Host:	Narcos-3.001_1 Host
Data Source:	Narcos-3.001
File:	/img_Narcos-3.001/vol_vol7/Users/JaneE/AppData/Local/Microsoft/Windows/WebCache/WebCacheV01.dat

Fig. 29 Web Search – “How to act like a meth addict”

Web Search	
Term:	how to pretend to be desperate in drug dealings
Time:	2019-01-31 23:07:06 GST
Domain:	bing.com
Program Name:	Microsoft Edge Analyzer
Source	
Host:	Narcos-3.001_1 Host
Data Source:	Narcos-3.001
File:	/img_Narcos-3.001/vol_vol7/Users/JaneE/AppData/Local/Microsoft/Windows/WebCache/WebCacheV01.dat

Fig. 30 Web Search – “How to pretend to be desperate in drug dealings”

Web Search	
Term:	methamphetamine crystal rock
Time:	2019-01-31 01:09:24 GST
Domain:	google.com.au
Program Name:	Microsoft Edge Analyzer
Source	
Host:	Narcos-3.001_1 Host
Data Source:	Narcos-3.001
File:	/img_Narcos-3.001/vol_vol7/Users/JaneE/AppData/Local/Microsoft/Windows/WebCache/WebCacheV01.dat

Fig. 31 Web Search – “Meth crystal rock”

Web Search	
Term:	passing drugs through nz customs
Time:	2019-01-31 01:08:13 GST
Domain:	bing.com
Program Name:	Microsoft Edge Analyzer
Source	
Host:	Narcos-3.001_1 Host
Data Source:	Narcos-3.001
File:	/img_Narcos-3.001/vol_vol7/Users/JaneE/AppData/Local/Microsoft/Windows/WebCache/WebCacheV01.dat

Fig. 32 Web Search – “Passing drugs through NZ customs”

Web Search	
Term:	quasar rat
Time:	2019-01-30 01:40:11 GST
Domain:	youtube.com
Program Name:	Microsoft Edge Analyzer
Source	
Host:	Narcos-3.001_1 Host
Data Source:	Narcos-3.001
File:	/img_Narcos-3.001/vol_vol7/Users/JaneE/AppData/Local/Microsoft/Windows/WebCache/WebCacheV01.dat

Fig. 33 Web Search – “Quasar rat”

Web Search	
Term:	federal agent badge au
Time:	2019-01-28 22:42:12 GST
Domain:	google.com.au
Program Name:	Microsoft Edge Analyzer
Source	
Host:	Narcos-3.001_1 Host
Data Source:	Narcos-3.001
File:	/img_Narcos-3.001/vol_vol7/Users/JaneE/AppData/Local/Microsoft/Windows/WebCache/WebCacheV01.dat

Fig. 34 Web Search – “Federal agent badge AU”

Web Search

Term: micro voice recorder
Time: 2019-01-29 00:05:25 GST
Domain: google.com.au
Program Name: Microsoft Edge Analyzer

Source

Host: Narcos-3.001_1 Host
Data Source: Narcos-3.001
File: /img_Narcos-3.001/vol_vol7/Users/JaneE/AppData/Local/Microsoft/Windows/WebCache/WebCacheV01.dat

Fig. 35 Web Search – “Micro voice recorder”

Visit Details

Username: JaneE
Date Accessed: 2019-01-29 00:06:42 GST
Domain: securitylab.com.au
URL: https://www.securitylab.com.au/spy-voice%20recorders/edic-mini-tiny-b73-listening-device-voice-recorder
Program Name: Microsoft Edge Analyzer

Source

Host: Narcos-3.001_1 Host
Data Source: Narcos-3.001
File: /img_Narcos-3.001/vol_vol7/Users/JaneE/AppData/Local/Microsoft/Windows/WebCache/WebCacheV01.dat

The screenshot shows the Security Lab website's product page for the 'Edic Mini Tiny+ B73 Spy Voice Recorder'. The page features a navigation bar with categories like SECURITY SYSTEM, SURVEILLANCE, and COUNTER SURVEILLANCE. The product is displayed with a large image of the black recorder and a red Coca-Cola bottle cap. The price is listed as \$499.00, with an 'afterpay' option available. The product is marked as 'IN STOCK' and 'Model: EDIC_MINI_B73'. There are buttons for 'ADD TO CART', 'BUY NOW', and 'Add to Wish List'. The page also includes a description and reviews section.

Fig. 36 Web Visit – “Spy Voice Recorder”

Visit Details

Username: JaneE
 Date Accessed: 2019-01-29 00:07:36 GST
 Domain: securitylab.com.au
 URL: https://www.securitylab.com.au/police-camera-body-video-recorder
 Program Name: Microsoft Edge Analyzer

Source

Host: Narcos-3.001_1 Host
 Data Source: Narcos-3.001
 File: /img_Narcos-3.001/vol_vol7/Users/JaneE/AppData/Local/Microsoft/Windows/WebCache/WebCacheV01.da



ALL Search here...

Logi

SECURITY SYSTEM SURVEILLANCE COUNTER SURVEILLANCE BODYCAMS DOSIMETERS GPS TRACKERS MEM

→ Bodycams

Bodycams



Filter

PRICE

\$ 180 \$ 400

X CLEAR

Body camera

Are you searching for small police bodycam? [Show More](#) widest selection of police cams in

Product Compare Sort By: Default Show: 12



Police Body Camera 2K GPS 4G

\$449.00



High-Grade 4G Police Body Camera 32MP



ULTRA HD 4K Portable Police Body Camera

Fig. 37 Web Visit – “Body camera recorder”

Visit Details

Username: JaneE
 Date Accessed: 2019-01-29 22:00:48 GST
 Domain: github.com
 URL: https://github.com/quasar/QuasarRAT/releases/download/v1.3.0.0/Quasar.v1.3.0.0.zip
 Program Name: Microsoft Edge Analyzer

Source

Host: Narcos-3.001_1 Host
 Data Source: Narcos-3.001
 File: /img_Narcos-3.001/vol_vol7/Users/JaneE/AppData/Local/Microsoft/Windows/WebCache/WebCacheV01.da

Fig. 38 Web Visit – “Install Quasar Remote Access Trojan”

3.10 Finding 10- Quasar RAT (Jane's Malware)

Path: /img_Narcos-3.001/vol_vol7/Users/JaneE/Downloads/Quasar
v1.3.0.0/Clients/JohnF@JOHNFLAPTOP1_4DD90B0/Logs/01-30-2019.html
Calculated Hash: fd3a044bee4d5af72faba9630867d4fd

Path: /img_Narcos-3.001/vol_vol7/Users/JaneE/Downloads/Quasar
v1.3.0.0/Clients/JohnF@JOHNFLAPTOP1_4DD90B0/Logs/01-31-2019.html
Calculated Hash: 948b4c559b944d6c0f2e1da8e285d835

Path: /img_Narcos-3.001/vol_vol7/Users/JaneE/Downloads/Quasar
v1.3.0.0/Clients/JohnF@JOHNFLAPTOP1_4DD90B0/Logs/02-01-2019.html
Calculated Hash: f51aa5754e54915e93d329a3eefdc042

Path: /img_Narcos-3.001/vol_vol7/Users/JaneE/Downloads/Quasar
v1.3.0.0/Clients/JohnF@JOHNFLAPTOP1_4DD90B0/Logs/02-02-2019.html
Calculated Hash: 7edea768c94997ed685bf7bc51cc325a

3.10.1 Discussion

During the investigation, it was revealed that Jane used a remote access trojan (RAT) known as "Quasar RAT." This tool, capable of spying and remotely controlling a target machine, was embedded in a file named "ContactCard.zip" that Jane sent to John via Discord. This allowed Jane to log his keystrokes and monitor his activities.

The keylogging feature of Quasar RAT provided Jane with critical information, including the decryption password for the "secret.txt" file and John's Protonmail credentials. This highlights Jane's efforts to gather information on John's illegal activities as part of her undercover assignment.

This evidence underscores Jane's role as an undercover agent, demonstrating that her actions were part of a legitimate operation to collect incriminating information on John.

Log created on 30.01.2019 08:03

[Cortana - 08:13]
power[Enter]

[#general - Discord - 08:14]
Sweet a[Back] [Back][Back] thanks[Enter]
Got a new supplier eh[Back][Back][Back] Ya Interested??[Enter]
Humm 10 is abit risky [Back][Back][Back][Back][Back][Back][Back][Back] bit much for the first time around and is pretty risky [Back]. How a[Back][Back][Back][Back]ow about we start off with 1 and can ramp up from there if a
Greta[Back][Back][Back]eat John out[Left][Right].[Left][Left][Left][Left][Right][Back]O[Left][Left][Left][Left][Left][Left][Left].[Enter]

[Start - Microsoft Edge - 09:56]
protonmail[Back][Back]l.com[Enter]

[Login | ProtonMail - Microsoft Edge - 09:56]
john[Back][Back][Back][Back][Back][Back]heresjohnny1John1234[Enter]

[4] Inbox | heresjohnny1@protonmail.com | ProtonMail - Microsoft Edge - 09:57
[Control + C]

[New tab and 1 more page - Microsoft Edge - 09:57]
[Control + V][Enter]

[Enter password for C:\Users\JohnF\Downloads\Attachments-Importa...secret - 10:04]
ilovediving

[Mozilla Firefox - 11:49]
decryption methods [Back]encryption methods youtube[Enter]

Log created on 31.01.2019 06:37

[Windows Default Lock Screen - 06:37]
[Up]

[Cortana - 06:38]
power set[Enter]

[#general - Discord - 09:43]
Yo, what you after [Back]?[Enter]
Also that contact card dod[Back][Back]idn't open and I can;t[Back][Back]t be bothered with new clients got bigger fishj to fry [Left][Left][Left][Left][Left][Left][Left][Left][Back][Right][Right][Right][Right][Right][Right][Right][Le
Aight cool, umm actually I have a proposition for ya[Enter]
I have a business dealingh [Back][Back] happenning overseas and I need you to accompany me as cover. Everything will be paid for but you need to keep your trap shut[Enter]
Yea I understand but ill make it worth your while[Enter]
I didn't [Back][Back][Back][Back]ja[Back]t want to do this but, you n[Back][Back] leave me no choice [Enter]
You don't wanna see them hurt do yvGood now that's more like it '[Back][Back][Back][Back][Back][Back][Back][Back], now that's what I wanted to hear [Left]. Here is the plan [Back]. YU[Back]you will [B
Good talk tomorrow [Left][Left][Left][Left][Left][Left][Left][Left][Left][Left][Left][Left][Left][Left][Right][Right][Back]T[Right][Right][Right][Right][Right][Right][Right][Right][Right][Right][Right][Right]at 3PM or els

[Mozilla Firefox - 11:59]
googl[Enter]

[Google - Mozilla Firefox - 11:59]
drug detect[Back][Back][Back][Back]tector dogs[Enter]

[drug detector dogs - Google Search - Mozilla Firefox - 12:01]
body packig[Back][Back]ing[Enter]

[body packing - Google Search - Mozilla Firefox - 12:04]
[Back]suitcase coil[Back]lucealments[Enter]

35

[body packing - Google Search - Mozilla Firefox - 12:04]
[Back]suitcase coi[Back]ncealments[Enter]

[suitcase concealments - Google Search - Mozilla Firefox - 12:06]
fro drugs[Enter]

[suitcase concealments fro drugs - Google Search - Mozilla Firefox - 12:13]
new zealnd [Back][Back][Back]and wellington [Enter]

[new zealand wellington - Google Search - Mozilla Firefox - 12:15]
[Back]flights to [Back][Back][Back][Back] brisbane to WLG return[Enter]

Fig. 41 Quasar RAT – “log 3”

Log created on 01.02.2019 07:02

[#general - Discord - 07:02]
ah bug

[#general - Discord - 07:02]
ger wrong person d[Back][Back], dsiregard[Back][Back][Back][Back][Back][Back][Back][Back]isreage[Back][Back][Back][Back]egard[Left][Right][Right][Left][Right] message[Back]Message[Back]I want to send an image into a [B
No what's that?[Enter]
Okay I'll [Back][Back][Back]ll [Left][Left][Left][Left][Left][Left][Right][Right][Right][Right][Right][Right]have a look and see if I can get it to work and then send the image through [Enter]
steganography

[Mozilla Firefox - 07:25]
stegna[Back][Back]o[Back]anography[Enter]

[steganography - Google Search - Mozilla Firefox - 07:29]
imaghe d[Back][Back][Back][Back]e download[Enter]

[steganography image download - Google Search - Mozilla Firefox - 07:29]
beach[Enter]

[beach - Google Search - Mozilla Firefox - 07:30]
risbane[Enter]

[brisbane - Google Search - Mozilla Firefox - 07:30]
tutorial on h[Back][Back]image stegnao[Back][Back][Back]anography[Enter]

[Mozilla Firefox - 08:59]
image steganogrpahy

Fig. 42 Quasar RAT – “log 4”

[Image steganography - Google Search - Mozilla Firefox - 08:59]
dvo[nl][Back][Back][Back]own[Down][Enter]

[Image Steganography - 09:02]
[Back][Right]

[Save As - 09:04]
BNE

[Enter Password - 09:05]
Elchapo2

[#general - Discord - 09:59]
It worked [Back], sending it via email now. I used a tool called image steganography [Left]. [Back][Back],[Right][Right][Right][Right][Right][Right][Right]and the password [Back][Back][Enter]

[Mozilla Firefox - 10:04]
em[Back][Back][Back]protonmail[Enter]

[Login | ProtonMail - Mozilla Firefox - 10:04]
crat[Back]yfi[Back]heresjohnny1@[Back]John1234

[(4) Inbox | heresjohnny1@protonmail.com | ProtonMail - Mozilla Firefox - 10:05]
crayfish1980@protstego imageimage attached.[Enter]
pass is Elchapo2

[#general - Discord - 11:58]
Right [Back]. Whats[Back]'s your full name and date of birth [Back]? I need it for booking flights [Left][Left][Left][Left][Left][Left][Left][Left] [Left][Left][Right]the[Right][Right][Right][Right][Right][Right][Right][Right]ase

[Program Manager - 13:13]
[Control + Alt + Delete]

Fig. 43 Quasar RAT – “log 5”

Log created on 02.02.2019 08:04

[Settings - 08:04]
computer

[Mozilla Firefox - 08:29]
flights from brisbane to wlg[Enter]

[flights from brisbane to wlg - Google Search - Mozilla Firefox - 08:29]
au[Enter]

[flights from brisbane to wlg au - Google Search - Mozilla Firefox - 08:34]
jetstar[Enter]

[flights from brisbane to wlg jetstar - Google Search - Mozilla Firefox - 08:40]
[Back][Back][Back][Back][Back][Back][Back][Back][Back][Back]ellington au[Enter]

[Cortana - 08:42]
snip[Enter]

[Save As - 08:42]
flight[Back][Back][Back][Back][Back]sty[Back][Back][Back]Steve K

[#general - Discord - 09:00]
Flights n[Back]booked [Back]. I'll pick you up from the Woolworths ([Left]133 Oxlet[Back]y Station Rd, Oxley QLD 4075[Back], Austraila[Right] at ... Just bring yourself t'I[Back]ll [Left][Left][Left][Back][Right][Right][Right][Dc
See you soon [Back]. John out.[Back]

[Mozilla Firefox - 12:40]
woolworths[Enter]

Fig. 44 Quasar RAT – “log 6”

[Woolworths Supermarket - Buy Groceries Online - Mozilla Firefox - 12:41]
travel[Enter]

[travel - Woolworths Online - Mozilla Firefox - 12:41]
suitcase[Enter]

[suitcase - Woolworths Online - Mozilla Firefox - 12:42]
[Back][Back][Back]case'[Back]\[Enter]

[case\ - Woolworths Online - Mozilla Firefox - 12:42]
[Right][Back][Enter]

[case - Woolworths Online - Mozilla Firefox - 12:42]
swim[Enter]

[swim - Woolworths Online - Mozilla Firefox - 12:42]
[Back]toilet[Enter]

[toilet - Woolworths Online - Mozilla Firefox - 12:43]
[Back]sanitary[Back][Back][Back][Back][Back]

[Dettol Foam Hand Wash Rose & Cherry In Bloom 250ml | Woolworths - Mozilla Firefox - 12:43]
[Back]toilet[Back][Back][Back][Back]it[Back][Back]bars[Enter]

[bars - Woolworths Online - Mozilla Firefox - 12:44]
[Enter]

Fig. 45 Quasar RAT – “log 7”

[Uncle Tobys Aussie Faves Muesli Bar Nanalicious 6 pack | Woolworths - Mozilla Firefox - 12:44]
[Back]es[Back]p

[Help At Hand Epsom Salt 1kg | Woolworths - Mozilla Firefox - 12:44]
[Back]sunblock[Enter]

[sunblock - Woolworths Online - Mozilla Firefox - 12:45]
tan[Enter]

[sunblock tan - Woolworths Online - Mozilla Firefox - 12:45]
[Back]directions from Inala to brisbane airport [Enter]

[Enter password for C:\Users\JohnF\Downloads\Attachments-Importa...\secret - 12:48]
ilovedivingi[Back]ilo

[TrueCrypt - Mount Options - 12:49]
ve

[Enter password for C:\Users\JohnF\Downloads\Attachments-Importa...\secret - 12:50]
ilovediving

Fig. 46 Quasar RAT – “log 8”

3.11 Finding 11 – Jane’s OneDrive

3.11.1 Discussion

After gaining access to Jane’s Protonmail, it was possible to access her OneDrive. The OneDrive contained a notebook that was Jane’s diary. This is a significant and decisive exculpatory discovery supporting Jane.

The notes revealed that Jane is an undercover agent who’s documenting her methods to blend in with meth users by using surveillance tools such as body cameras, voice recorders, and Quasar malware. The notes also covered her interactions with John and how she was trying her best to maintain her cover. This evidence substantiates her innocence and highlights her crucial role in the investigation.

3.11.2 Supporting Evidence

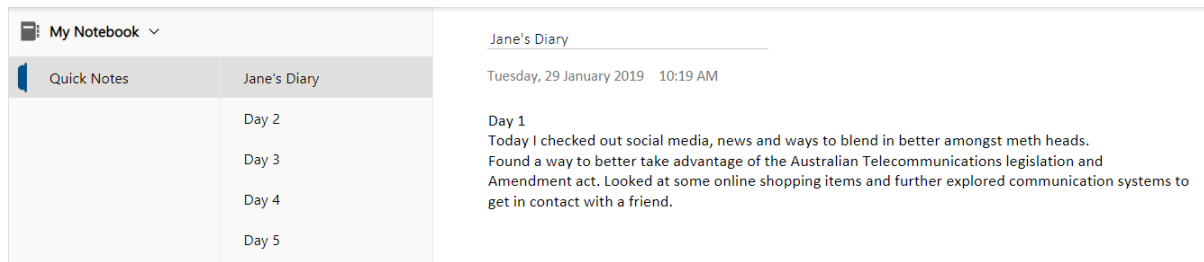


Fig. 47 OneDrive – “Jane’s Notes Day 1”

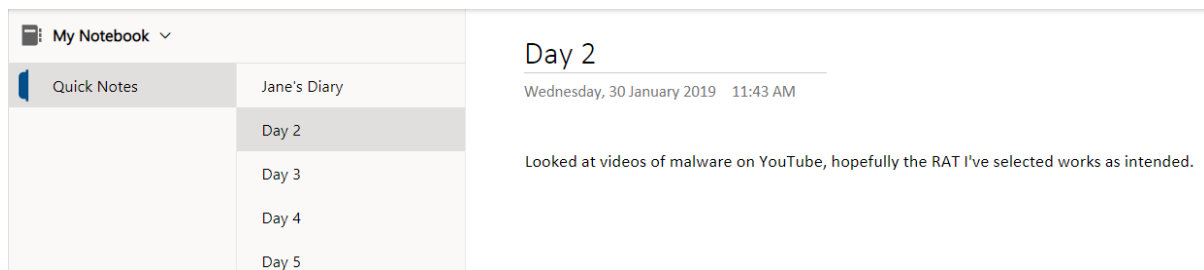


Fig. 48 OneDrive – “Jane’s Notes Day 2”

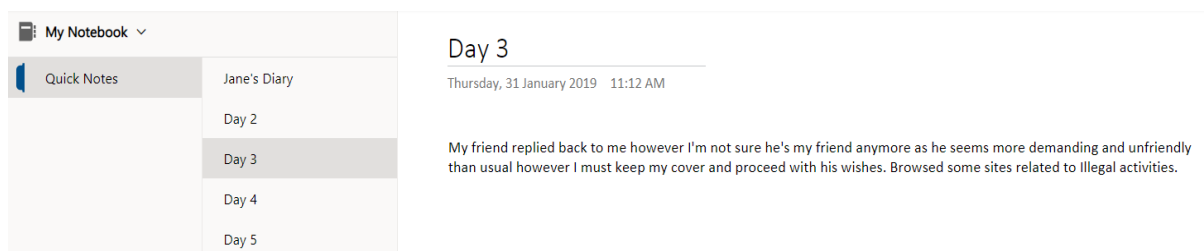


Fig. 49 OneDrive – “Jane’s Notes Day 3”

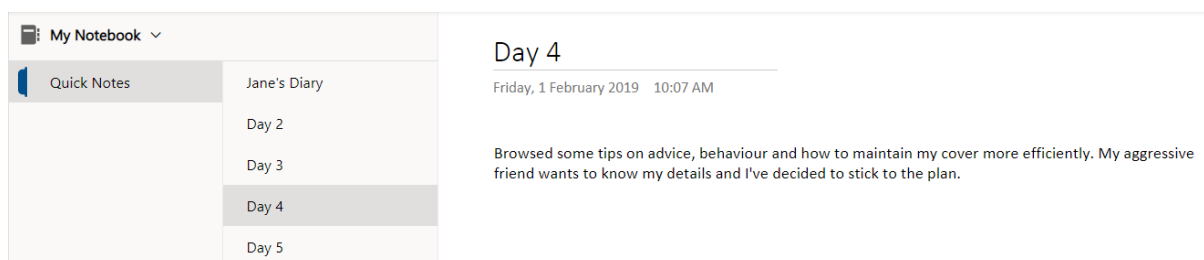


Fig. 50 OneDrive – “Jane’s Notes Day 4”

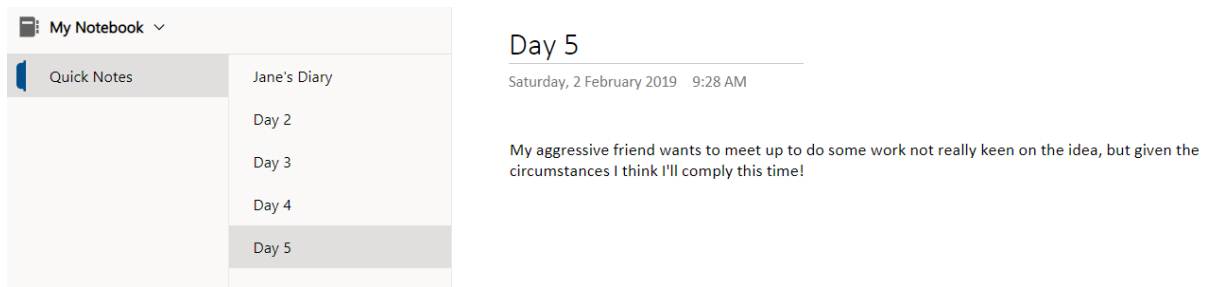


Fig. 51 OneDrive – “Jane’s Notes Day 5”

3.12 Finding 12- Discord Chat Communications (Jane & John)

Path:/Users/JohnF/AppData/Roaming/Discord/Local Storage/leveldb/000013.log

3.12.1 Discussion

The analysis of Discord communications reveals critical evidence exculpatory to Jane and incriminating to John. The chat records show John’s coercive behavior, where he blackmailed Jane by sending a picture of her children to force her into participating in a drug-related trip. Initially, Jane contacted John through Discord to purchase drugs and shared a file, "Contact_Card.zip," containing RAT malware. John subsequently invited Jane to join him overseas for a business deal. When Jane declined, he used the threat against her children to compel her compliance.

This evidence clearly demonstrates Jane’s coerced involvement, proving her actions were driven by threats rather than voluntary participation, thereby exonerating her. Conversely, it highlights John’s active role in the illegal operation, underscoring his culpability in the drug trafficking scheme.

3.12.2 Supporting Evidence

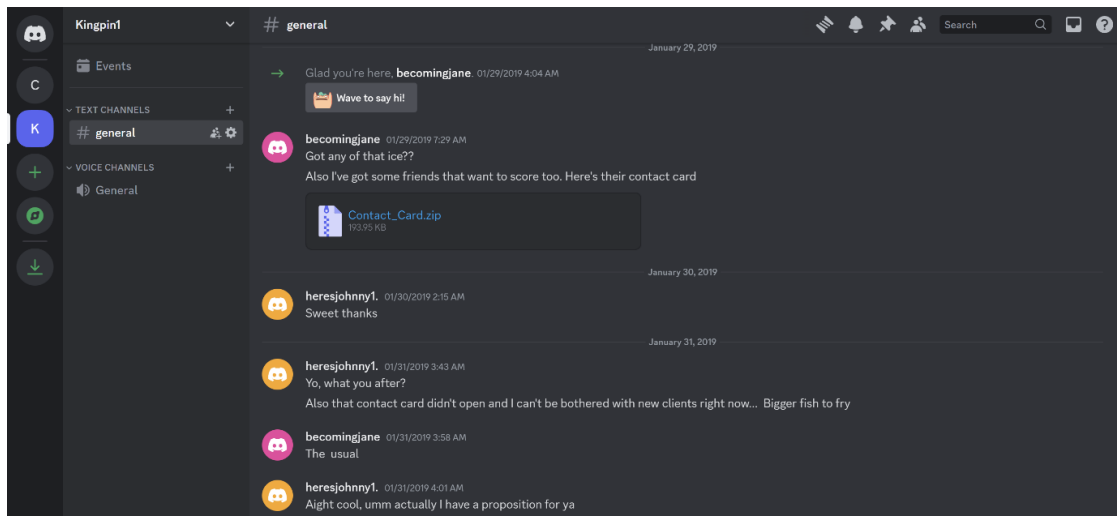


Fig. 52 Discord Chats (1/4) – “Jane & John Discord Communication”

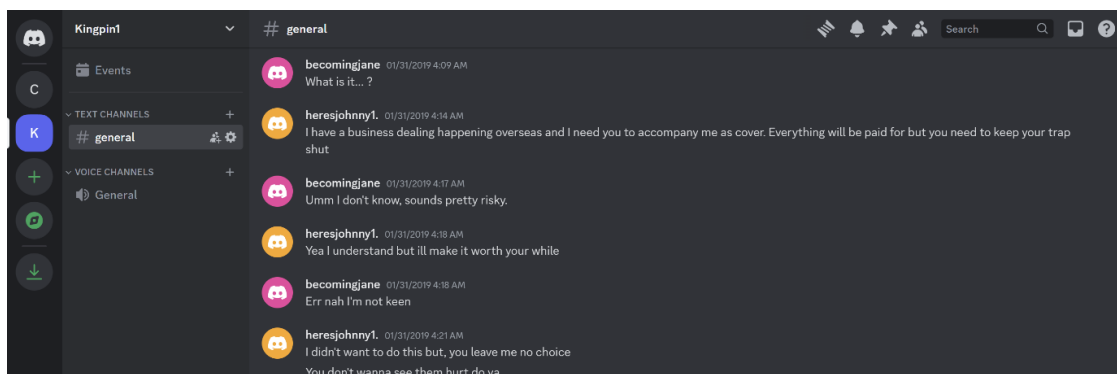


Fig. 53 Discord Chats (2/4) – “Jane & John Discord Communication”

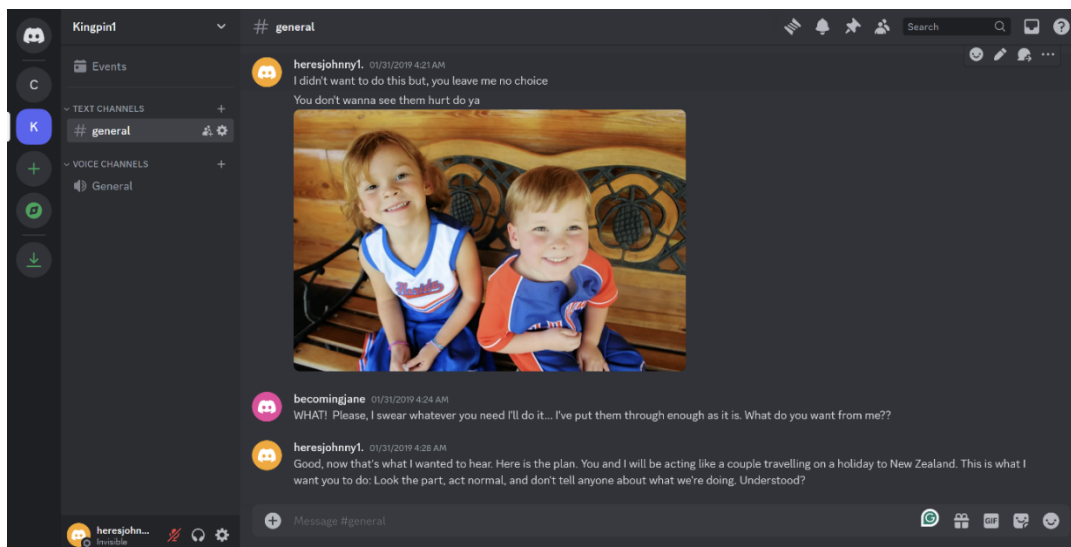


Fig. 54 Discord Chats (3/4) – “Jane & John Discord Communication”

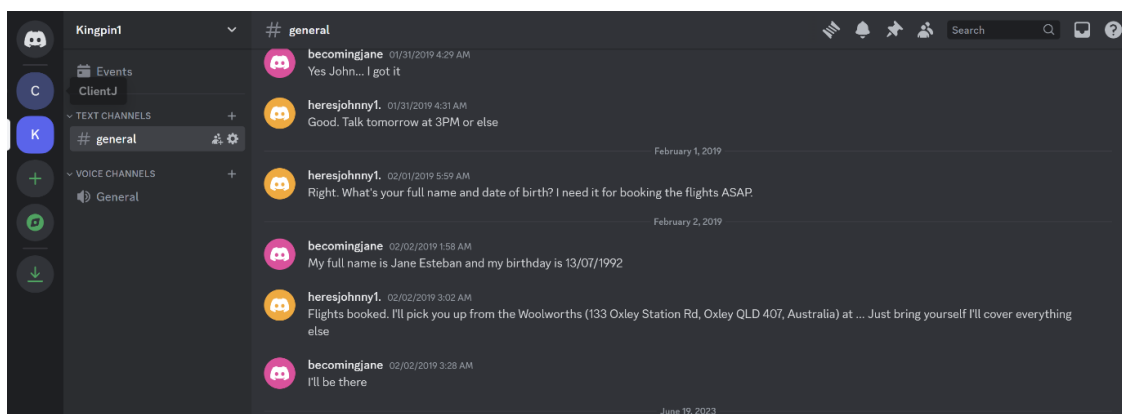


Fig. 55 Discord Chats (4/4) – “Jane & John Discord Communication”

3.13 Finding 13 – Jane Drug Images

Path: /img_Narcos-3.001/vol_vol7/Users/JaneE/Pictures/crys1.jpg

Calculated Hash: 5c580e37403cc6138497a23e9b0b9464

Path: /img_Narcos-3.001/vol_vol7/Users/JaneE/Pictures/crys3.jpg

Calculated Hash: 4eb61426baf813858640533599816096

Path: /img_Narcos-3.001/vol_vol7/Users/JaneE/Pictures/crys2.jpg

Calculated Hash: 9d005f3905b24b628f5eb076c8524bcf

3.13.1 Discussion

Several drug-related images were discovered in Jane's files. Given that Jane was an undercover officer, these images were maintained as part of her effort to convincingly portray the persona of a drug dealer with an addiction.

3.13.2 Supporting Evidence



Fig. 56 Drug Samples – “Jane’s Drugs 1”



Fig. 57 Drug Samples – “Jane’s Drugs 2”



Fig. 58 Drug Samples – “Jane’s Drugs 2”

4. Summary & Conclusion

4.1 Illegal Drug Activity Evidence

Following a comprehensive review, evidence of illicit drug activities involving methamphetamine was uncovered. Steve’s files included images of the drugs he was distributing and requests made to John, while John’s files listed his regular clients who purchased methamphetamine and other drugs from him.

An encrypted document between John and Steve outlined their plans for smuggling drugs into New Zealand. Furthermore, an image encoded with steganography unveiled a concealed drug package that John was transporting for Steve.

4.2 Reliability of Evidence

The evidence collection adhered to rigorous forensic protocols, including hashing, to ensure the accuracy and integrity of the findings. Each step was meticulously documented and managed according to established forensic procedures, guaranteeing that the evidence remained intact and dependable throughout the investigation. Furthermore, there is ample evidence available for additional review and verification to reconfirm its reliability.

4.3 Verdict

John Fredericksen and Steve Kowhai have been implicated in illegal drug activities, including the distribution and smuggling of methamphetamine. The evidence against them is substantial, comprising drug-related images, encrypted documents detailing their smuggling operations, steganographic images revealing hidden drug packages, and incriminating communications via Discord chats. Additionally, web searches conducted by them further corroborate their involvement in these criminal activities.

Conversely, Jane Esteban has been exculpated, as she was an undercover officer with the Australian Federal Police. Her role involved assuming the identity of a drug addict to collect evidence on Fredericksen and Kowhai's illicit operations. Evidence supporting her innocence includes her official police badge, web search history showing her investigative activities, records of John's coercive blackmail tactics, and the use of Quasar RAT to monitor John's activities. This evidence collectively affirms her genuine role as an undercover agent and her lack of involvement in drug-related crimes.

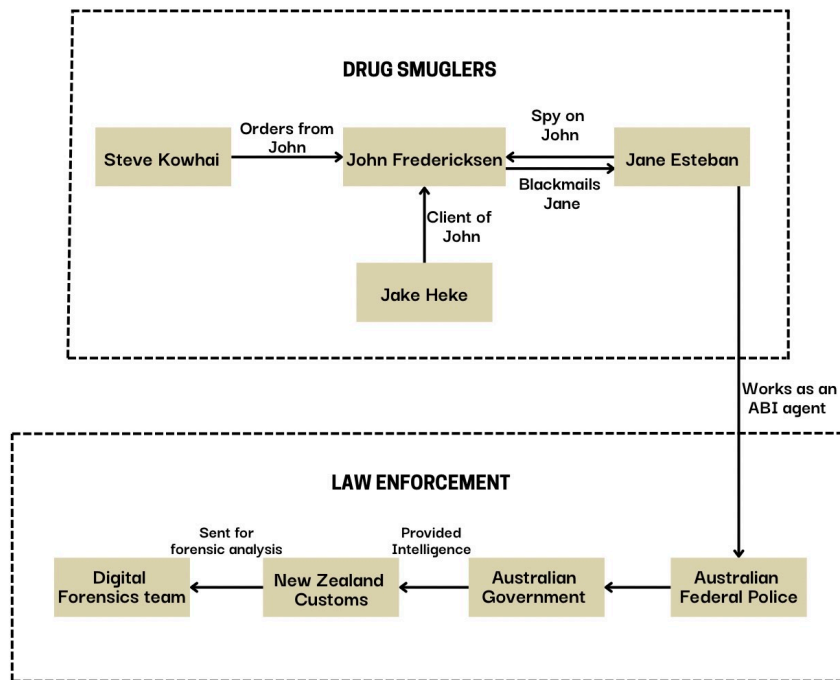
5. Appendix

5.1 Persons of Interest

Name	Interest
Steve Kowhai	Steve Kowhai, a prominent drug dealer in New Zealand, is notorious for his extensive operations in the southern region. In his efforts to expand his drug empire, Kowhai arranged to acquire test samples of methamphetamine from John Fredericksen. Kowhai, as the principal buyer, engages directly with Fredericksen to evaluate and purchase methamphetamine, which he assesses for quality. Their communication is conducted through encrypted channels, specifically ProtonMail, and involves the exchange of encrypted documents to maintain secrecy. Kowhai provided Fredericksen with

	detailed instructions on how to discreetly smuggle the new drug samples from the United States into Wellington
Jane Esteban	Jane Esteban, an undercover officer with the Australian Federal Police, was assigned to infiltrate a drug ring by posing as a methamphetamine user. Her primary mission was to gather intelligence on the drug operation's activities. As a key operative, Jane successfully maintained her cover while collecting crucial evidence and identifying significant figures within the network. Her efforts played a pivotal role in exposing the operation, ultimately leading to John Fredericksen's capture.
John Fredericksen	John Fredericksen, an international drug dealer based in the United States, plays a central role in smuggling drugs into New Zealand. Fredericksen, who maintains a global client base and continually seeks to expand his operations, arranged to deliver a 1 kg sample of methamphetamine to Steve Kowhai. To facilitate this operation, Fredericksen coordinated with Kowhai for the delivery and involved Jane Esteban as a cover agent to ensure his escape if needed. The plan included Fredericksen traveling by plane with Esteban to Wellington, with specific meeting locations.
Jake Heke	Jake Heke is a client of John, that resides in new Zealand. Jake has ordered a shipment of 10kg of tweak to be delivered using discord.
New Zealand Authorities	The New Zealand authorities, acting on intelligence provided by the Australian Federal Police, disrupted the drug operation and captured Jane and John from the airport. They are currently engaged in monitoring and investigating the case, and are responsible for conducting digital forensics on the devices of the three suspects.

5.2 Association Diagram of Persons of Interest



5.3 Evidence Listing

Evidence Number	File Type	Name	Creation Date	File Path
1	SQLite	John's online activities and interactions	29-01-2019 to 31-01-2019	Path:Users/JohnF/AppData/Roaming/Mozilla/Firefox/Profiles/fj248tbu.default/places.sqlite
2	DAT	Steve's online activities and interactions	28-01-2019 - 01-02-2019	Path: /Users/Steve/AppData/Local/Microsoft/Windows/WebCache/WebCacheV01.dat
3	ODS	Evidence of Established Drug Distribution Network	29-01-2019	Path:/Users/JohnF/Documents/Business/clients.ods

4	JPG	Image Stenography Communication (Steve & John)	2019-02-01	FILENAME: BNE.PNG Path: /Users/Steve/Downloads/Misc/BNE.png FILENAME: PACKAGE.JPG Path:/Users/JohnF/AppData/Local/Temp/v mware- JohnF/VMwareDnD/770cd05f/package.jpg
5	TXT	Document Stenography Communication (Steve & John)	29-01-2019	Path:/Users/JohnF/Downloads/Attachments-Important, crucial to our method/secret
6	Discord Logs and PNG	Discord Chat Communications (Steve & John)	2019-01-29 to 2019-02-02	LOG NAME: 000006.log Path: /Users/Steve/AppData/Roaming/Discord/L ocal Storage/leveldb/000006.log IMAGE NAME: FLIGHTBOOKINGS.PNG Path: /Users/Steve/Documents/Misc/flightbookings.PNG
7	JPG	Drug Samples	2019-01-31	Path: /\$Recycle.Bin/S-1-5-21-1474204758- 2504895174-1356074821- 1001/\$RA3IE5E.jpg Path: /\$Recycle.Bin/S-1-5-21-1474204758- 2504895174-1356074821- 1001/\$RIIK1AS.jpg

8	JPG	Drug meetup location (Steve & John)	2019-01-31 to 2019-02-02	Path: /Users/Steve/Documents/Misc/airport crystals.jpg Path: /Users/Steve/Documents/Misc/dropoff.jpg
9	DAT	Jane's online activities and interactions	29-01-2019 to 1-02-2019	Path: /Users/JaneE/AppData/Local/Microsoft/Windows/WebCache/WebCacheV01.dat
10	HTML	Quasar RAT (Jane's Malware)	01-30-2019 to 02-02-2019	Path: /img_Narcos-3.001/vol_vol7/Users/JaneE/Downloads/Quasar v1.3.0.0/Clients/JohnF@JOHNFLAPTOP1_4 DD90B0/Logs/01-30-2019.html Path: /img_Narcos-3.001/vol_vol7/Users/JaneE/Downloads/Quasar v1.3.0.0/Clients/JohnF@JOHNFLAPTOP1_4 DD90B0/Logs/01-31-2019.html Path: /img_Narcos-3.001/vol_vol7/Users/JaneE/Downloads/Quasar v1.3.0.0/Clients/JohnF@JOHNFLAPTOP1_4 DD90B0/Logs/02-01-2019.html Path: /img_Narcos-3.001/vol_vol7/Users/JaneE/Downloads/Quasar v1.3.0.0/Clients/JohnF@JOHNFLAPTOP1_4 DD90B0/Logs/02-02-2019.html

				uasar v1.3.0.0/Clients/JohnF@JOHNFLAPTOP1_4 DD90B0/Logs/02-02-2019.html
11	OneDrive	Jane's OneDrive	01-29-2019 to 02-02-2019	Jane Notes on her OneDrive Account
12	Discord Log	Discord Chat Communications (Jane & John)	29-01-2019 to 31-01-2019	Path:/Users/JohnF/AppData/Roaming/Discord/Local Storage/leveldb/000013.log
13	JPG	Jane Drug Images		Path: /img_Narcos-3.001/vol_vol7/Users/JaneE/Pictures/cry1.jpg Path: /img_Narcos-3.001/vol_vol7/Users/JaneE/Pictures/cry2.jpg Path: /img_Narcos-3.001/vol_vol7/Users/JaneE/Pictures/cry3.jpg

5.4 Evidence Timeline

Number	Evidence	TimeLine	Brief Description
1	shipping.PNG	Created: 2019-01-23 05:18:28 GST	A package is intended to be shipped online to the John Heke address.

2	Steve's suspicious web search starts	Created: 2019-01-28 TO 2019-02-02	Steve was involved in illegal activities such as drug smuggling and money laundering during the reported time frame.
3	Jane Undercover web-related search starts	Created: 2019-01-28 TO 2019-02-02	Jane spent time looking for ways to maintain her cover as a meth addict and sought spying equipment and malware.
4	John's suspicious web search starts	Created: 2019-01-29 TO 2019-02-02	John's online activities included illegal actions like drug smuggling and breaching security in NZ, as detailed in the report.
5	John's client list file(Clients.ods) file is created	Created: 2019-01-29 00:26:09 GST	A file was generated detailing John's international clients, the types of drugs they receive, and the quantities.
6	Steve and John Discord chat starts	2019-01-29 01:17 AM TO 2019-02-02 01:17 AM	Steve uses Discord to contact John about obtaining 10,000 drugs and discusses methods for securely sharing documents.
7	John and Jake Heke's Discord chat starts	Created: 2019-01-29 06:14 AM TO 2019-01-29 06:20 AM	Jake, a client of John, inquires about the shipment of 10 kilograms of drugs.

8	John and Jane Estaban's Discord chat starts	Created: 2019-01-29 07:29 AM TO 2019-02-02 03:28 AM	John asks Jane to accompany him to New Zealand.
9	Jane sends the(Contact_Card.zip) Quasar RAT malware to John	Created: 2019-01-29 07:29 AM	Jane sends Quasar a RAT disguised as a contact list for clients needing John's services.
10	Jane started writing notes on OneDrive	Created: 2019-01-29 10:19 AM TO 2019-02-02 09:28 AM	Jane begins documenting key events and activities related to her undercover work, noting her actions and future plans.
11	QUASAR Malware creates logs	Created: 2019-01-30 3:08 AM TO 2019-02-02 08:04 AM	The malware Jane sent to John was activated, generating logs with sensitive information from John's PC, including his email password, secret documents, and hidden images.
12	Steve sends John the secret document	Created: 2019-01-30 3:40 AM	Steve transmits a secret file encrypted with TrueCrypt, which contains operational details and a "memo things" Word document after decryption.
13	John threatens Jane's kids on discord	Created: 2019-01-31 4:11 AM	John threatens Jane with harm to her children unless

			she agrees to go with him to New Zealand.
14	John sends Steve a hidden image in the BNE.png image. The hidden image is package.jpg.	Created: 2019-02-01 4:11 AM	John conceals an image of a 1-kilogram drug package within a (BNE.png) image using a steganography tool.

5.5 Software and Tools used in the investigation

Autopsy v-4.21.0

Purpose: Conduct a forensic investigation of the suspects' virtual images.

Functionality: Autopsy was used to add the image as a data source for a new case. Its powerful variety of tools and ingest modules allowed for a thorough exploration of file systems, extraction and analysis of metadata, recovery of deleted files, and detection of user behavior. Keywords were searched to obtain relevant data.

Strengths:

- Open-source and free to use with a large community of developers.
- Simple user interface suitable for both beginners and experienced investigators.
- Supports a wide range of file systems and file types for comprehensive analysis.

Weaknesses:

- Can be slow when analyzing large datasets, especially on older technology.

OSForensics v-11.0

Purpose: Analyze Steve's and other suspects virtual image comprehensively.

Functionality: OSForensics' advanced features were used to scan and index the file system, search for specific file types, examine metadata, and recover deleted or hidden files. Its powerful search capabilities identified and extracted critical evidence by methodically analyzing the data and correlating it with the investigation objectives.

Strengths:

- Simple interface with workflow guides for standard forensic tasks.
- Offers various forensic analysis features, including disk imaging, file system analysis, keyword searching, and memory analysis.
- Built-in tools for file hashing, timeline analysis, email analysis, and browser history analysis.

Weaknesses:

- Commercial software may require a license for full operation and upgrades.
- Primarily designed for Windows systems, with limited support for non-Windows systems.

WinHex v-21.2

Purpose: Thorough examination of memory dumps from the suspects' virtual images.

Functionality: Loaded memory dump files and used WinHex's powerful hex editor to navigate through raw data. Analyzed memory contents, active processes, loaded modules, and network connections at the time of memory capture. Searched for specific keywords, file signatures, and other indicators of suspicious activity.

Strengths:

- Offers dual views (hexadecimal and ASCII) for flexible data examination and editing.
- Contains tools for analyzing disk structures, finding specific data patterns, and inspecting volatile memory contents.

- Enables data carving techniques for recovering fragmented or lost files using file signatures.

Weaknesses:

- User interface may not be intuitive, and there are limitations regarding viewing text.

Image Steganography v-1.5.2.0

Purpose: Detect and decode hidden data within images.

Functionality: It was used in this investigation because John and Steve were found to have hidden secret information in images. This tool was specifically used to conceal a hidden image containing crucial details about their operation.

Strengths:

- Simple interface suitable for all users.
- Compatible with various image formats.
- Basic encryption offers some security for hidden messages.
- Free of cost.

Weaknesses:

- Lacks strong algorithms and improved data processing.
- Update issues and size restrictions.

TrueCrypt v-7.2

Purpose: Decrypt encrypted files and drives.

Functionality: A very famous encryption software for volumes and files, it was used by the two suspects, John and Steve, to encrypt an important document regarding their drug operation.

Strengths:

- Uses strong encryption algorithms like AES, Serpent, and Twofish.
- Encrypts data in real-time while retaining functionality.
- Cross-platform availability (Windows, macOS, Linux).
- Free and open-source.

Weaknesses:

- Development ceased in 2014, potentially leading to security issues.
- May be difficult to configure for non-technical users.
- Compatibility issues with newer operating systems.
- Encryption can slow down system performance.

MD5 Calculator

Purpose: Ensure the integrity of files and evidence.

Functionality: Calculated MD5 hashes of files, ISO images, and memory dumps to check for any modifications during the investigation.

Strengths:

- Efficient checksum verification for file integrity.
- Quick calculation capabilities.
- Simple user interface.

Weaknesses:

- Relies on a single hashing algorithm (MD5).
- Depends on initial file integrity checks without continuous protection against tampering.

5.6 Other Listings

5.6.1 ProtonMail Credentials

Name	Username	Password
Steve	crayfish1980@protonmail.com	Steve1234
Jane	becomingjane@protonmail.com	Jane1234
John	heresjohnny1@protonmail.com	John1234

5.6.2 Discord Credentials

Name	Username	Password
Steve	crayfish1980@protonmail.com	Steve1234@
Jane	becomingjane@protonmail.com	Jane1234@
John	heresjohnny1@protonmail.com	JOHNF@1234

5.6.3 File Credentials

Encrypted File Name	Decrypted File Name	Password
BNE.png	Package.jpg	Elchapo2
Secret.txt	Memo Things	ilovediving

5.6.4 References

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